COMPUTER NETWORKS: NUMERICAL & FORMULA MASTER GUIDE

A Complete Pre-Exam Cheat-Sheet for Calculations, Bit Layouts, Formulas, and Shortcuts (Lectures 01 – 10)

1. Physical Topology Calculations & Formulas

Network topologies define physical or logical arrangements. Exam questions focus on physical link counts, port requirements per device, and total system hardware ports:

Topology	Physical Links Formula	Ports per Device	Total System Ports	Exam Key / Shortcut
Mesh (Duplex)	**`L = n(n-1) / 2`**	**`P = n - 1`**	**`P_total = n(n-1)`**	Every device has dedicated link to every other device.
Mesh (Simplex)	**`L = n(n-1)`**	**`P = 2(n - 1)`**	**`P_total = 2n(n-1)`**	Separate directional link for transmit and receive.
Star	**`L = n`**	1 port (to Hub/Switch)	**`P_hub = n`**	Devices connect directly to central Hub/Switch.
Bus	1 main cable + `n` drop lines	1 tap per node	`n` Taps + 2 Terminators	Requires 2 terminators at ends + BNC connectors.
Ring	**`L = n`**	2 ports (in & out)	**`P_total = 2n`**	Token passing; 1 repeater per node.
Tree	**`L = n - 1`**	1 port (upward)	Depends on switch hierarchy	Hubs/Switches connected in hierarchical branching.

FORMULA & SHORTCUT: MESH TOPOLOGY SHORTCUT TRICK

- **For `n` devices in a fully connected True Mesh:**
- **Direct links = `n × (n - 1) / 2`. (Example: 8 nodes → 8 × 7 / 2 = 28 links).**
- **Physical ports on EACH node = `n - 1`. (Example: 8 nodes → 7 ports per node).**
- **Quick sanity check: Total ports across all nodes = `2 × Total Links`.**

2. Addressing Levels, Bit Dimensions & Field Layouts

Networking uses 4 distinct addressing levels. Memorize these exact bit lengths, hexadecimal formats, and layer scopes for quick points:

Address Type	Bit Length	Byte Length	Layer Scope	Format Example & Notation
Physical (MAC)	48 bits	6 bytes	Data Link Layer (L2)	`00:1A:2B:3C:4D:5E` (12 Hex digits)
Logical (IPv4)	32 bits	4 bytes	Network Layer (L3)	`192.168.1.1` (4 Dotted-decimals)
Logical (IPv6)	128 bits	16 bytes	Network Layer (L3)	**`2001:0db8:85a3::8a2e:0370:7334`**
Port Address	16 bits	2 bytes	Transport Layer (L4)	`Port 80` (HTTP), `443` (HTTPS), `21` (FTP)
Stat-TDM Header	**`n = ceil(log2 N)`**	Var bits	Multiplexing Layer	`3 bits` for 8 output lines; `4 bits` for 16 lines

FORMULA & SHORTCUT: ADDRESS SPACE & PORT IDENTIFICATION RULES

- **IPv4 Total Address Space = 2^32 ≈ 4.3 Billion addresses.**
- **MAC Address Structure: First 24 bits = OUI (Vendor ID assigned by IEEE); Last 24 bits = Unique NIC serial number.**
- **Socket Address = `IP Address + Port Number` (e.g., `192.168.1.10:80`). Uniquely identifies a specific app on a host.**
- **Statistical TDM Address Bit Formula: To differentiate `N` output channels, slot address bits = `ceil(log2 N)`.**

3. Multiplexing Mathematics & Channel Formulas

Multiplexing combines multiple input signals over a single high-capacity link. Calculations fall into three primary categories: FDM Bandwidth, Synchronous TDM Timing, and Data Rate Equalization.

A. Frequency-Division Multiplexing (FDM) Calculations

FORMULA & SHORTCUT: FDM TOTAL BANDWIDTH FORMULA

- **For `N` voice/data channels of bandwidth `B_channel` with guard band `B_gb` between adjacent channels:**
- **Minimum Link Bandwidth: `B_total = (N × B_channel) + ((N - 1) × B_gb)`**
- **Golden Rule: `N` channels require exactly `(N - 1)` guard bands!**
- **Example Problem: 7 channels of 200 kHz with 20 kHz guard bands:**
- **-> `B_total = (7 × 200 kHz) + (6 × 20 kHz) = 1400 kHz + 120 kHz = 1520 kHz`.**

B. Synchronous TDM Time Slot & Frame Mathematics

Parameter	Symbol / Formula	Physical Meaning & Calculation Step
Input Bit Duration	**`T_in = 1 / R_in`**	Time required to send 1 bit on an input line.
Output Slot Duration	**`T_slot = T_in / n`**	Duration of a single slot on output link (`n` = input lines).
Frame Duration	**`T_frame = T_in`**	Duration of 1 complete frame = duration of 1 input bit.
Link Data Rate	**`R_link = n × R_in`**	Aggregate bit rate of the combined output link.
Frame Rate	**`F = R_in`**	Number of frames sent per second = Input data rate (for 1 bit/slot).
Link Bit Rate (with framing)	**`R_link = F × Bits_per_frame`**	Bits_per_frame = `(n × bits_per_slot) + framing_bits`.

FORMULA & SHORTCUT: SYNCHRONOUS TDM WORKED EXAMPLE

- **Problem: 4 input lines of 20 kbps each. Each slot carries 1 bit. 1 framing bit added per frame.**
- **1. Input bit duration `T_in = 1 / 20,000 s = 50 μs`.**
- **2. Output slot duration `T_slot = T_in / 4 = 50 μs / 4 = 12.5 μs`.**
- **3. Frame Rate `F = 20,000 frames/sec` (since 1 bit/slot from 20 kbps line).**
- **4. Bits per frame = `(4 × 1 bit) + 1 framing bit = 5 bits/frame`.**
- **5. Total Link Rate = `20,000 frames/s × 5 bits/frame = 100 kbps`.**

C. Data Rate Equalization Strategies in TDM

Equalization Strategy	When Applied?	Mathematical Operation / Shortcut Rule
Multilevel Multiplexing	Input rates are integer multiples.	Combine lower-speed lines first using sub-MUX: `2 × 20 kbps = 40 kbps`, then feed into main MUX.
Multiple-Slot Allocation	One line is much faster than others.	Split high-speed line with Serial-to-Parallel converter: `50 kbps` split into `2 × 25 kbps` slots.
Pulse Stuffing (Bit Padding)	Input rates are non-integer multiples.	Raise all lines to dominant rate `R_max` by adding dummy bits: `R_stuffed = R_max - R_actual`.

4. Wireless Propagation, Frequencies & Heights

Unguided transmission media transport electromagnetic waves without physical conductors. Review these exact frequency ranges, distances, and propagation thresholds:

Propagation Method	Frequency Range	Behavior & Mechanics	Applications
Ground Wave	**`Below 2 MHz`**	Follows Earth's curvature; hugs ground; omnidirectional.	AM radio, VLF/LF navigation locators.
Sky Wave	**`2 MHz to 30 MHz`**	Bounces / reflects off atmospheric ionosphere.	Citizens Band (CB) radio, shortwave, international radio.
Line-of-Sight (LOS)	**`Above 30 MHz`**	Straight-line direct antenna-to-antenna beam.	FM radio, TV, Cellular, Satellite, Terrestrial Microwave.

Wireless Band	Frequency Spectrum	Antenna / Wave Nature	Key Applications / Limitations
Radio Waves	**`3 kHz to 1 GHz`**	Omnidirectional (all directions)	AM/FM radio, TV, cordless phones; penetrates walls.
Microwaves	**`1 GHz to 300 GHz`**	Unidirectional (narrow beam)	Cellular, Satellite, Wireless LAN; parabolic dish focus.
Infrared (IR)	**`300 GHz to 400 THz`**	Line-of-sight short range	TV remotes, IrDA ports (4 Mbps); CANNOT penetrate walls.

FORMULA & SHORTCUT: SATELLITE COMMUNICATIONS REFERENCE CONSTANTS

- **`Geostationary Orbit Height: Exactly 22,000 miles (~36,000 km) above Earth's equator.`**
- **`Round-Trip Propagation Delay: ~280 milliseconds (due to vast distance to/from satellite).`**
- **`Satellite Role: Acts as an ultra-high altitude line-of-sight repeater / frequency transponder.`**

5. Data Communication Units, Conversions & Frame Math

Mastering data unit conversions and Ethernet frame structures is essential for numerical problems:

Unit / Parameter	Conversion / Bit Equivalent	Exam Formula / Context
1 Byte (Octet)	8 bits	Standard memory unit for framing and packet headers.
Data Rate Multipliers	**`1 kbps = 1,000 bps`** **`1 Mbps = 1,000,000 bps`** **`1 Gbps = 1,000,000,000 bps**	Networking uses decimal powers of 10 (`10^3`, `10^6`, `10^9`) for data rates!
Bit Duration (`T_b`)	**`T_b = 1 / Bit_Rate`**	Time to send 1 bit. Example: `1 Mbps -> T_b = 1 μs`.
Ethernet MTU Payload	46 to 1,500 bytes	Data payload field. Stuffed if < 46 bytes; fragmented if > 1500.
Ethernet Header overhead	18 bytes (14 header + 4 CRC)	6-byte Dest MAC + 6-byte Src MAC + 2-byte Type + 4-byte CRC.
Ethernet Frame Range	64 bytes to 1,518 bytes	Minimum frame size = 64B (46+18); Maximum = 1518B (1500+18).

6. Master Exam Formula & Calculation Summary

Keep this single summary table in front of you during final pre-exam revision:

Topic / Problem Type	Formula / Rule	Key Variable Definitions	Instant Shortcut Rule
Mesh Links (Duplex)	`L = n(n - 1) / 2`	`n` = number of nodes	Multiply `n` by `(n-1)` and divide by 2.
Mesh Ports	`P = n - 1`	`P` = ports per device	Each node needs 1 port per neighbor.
Star Links	`L = n`	`n` = attached hosts	1 dedicated link per host to switch.
FDM Bandwidth	`B = N(B_ch) + (N-1)B_gb`	`N` = channels, `B_gb` = guard	`N` channels need `(N-1)` guard bands!
TDM Slot Time	`T_slot = T_in / n`	`n` = number of inputs	Slot time shrinks by factor of `n`.
TDM Link Speed	`R_link = n × R_in`	`R_in` = input rate	Link speed is `n` times input channel speed.
TDM Frame Rate	`F = R_in`	Assuming 1 bit/slot	Frame rate equals single input line rate.
Stat-TDM Header	`bits = ceil(log2 N)`	`N` = output lines	2^3=8 -> 3 bits; 2^4=16 -> 4 bits.
MAC Address	48 bits = 6 bytes	Layer 2 Physical address	12 Hex digits (`FF:FF:FF:FF:FF:FF`).
IPv4 Address	32 bits = 4 bytes	Layer 3 Logical address	4 Dotted decimal numbers.
IPv6 Address	128 bits = 16 bytes	Layer 3 Logical address	8 Hex blocks separated by colons.
Port Address	16 bits = 2 bytes	Layer 4 Process address	0 to 65,535 (HTTP=80, HTTPS=443).
Ground Wave	`< 2 MHz`	Low frequency	Hugs Earth's surface.
Sky Wave	`2 MHz to 30 MHz`	Medium / High frequency	Bounces off ionosphere.
Line-of-Sight	`> 30 MHz`	Very High frequency	Requires direct antenna alignment.

FORMULA & SHORTCUT: EXAM TIP FOR NUMERICAL QUESTIONS

• 1. ALWAYS write down the formula first (e.g., `L = n(n-1)/2` or `B_total = N(B_ch) + (N-1)B_gb`). Professors give partial credit!
• 2. Double check units! Convert kbps -> bps, MHz -> kHz, and Bytes -> bits before calculating.
• 3. State variable definitions clearly (e.g., Let `n = 10` devices, `N = 7` channels).